

Process Mining:

Discovering processes from event logs

BPM

Cse345

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Outline

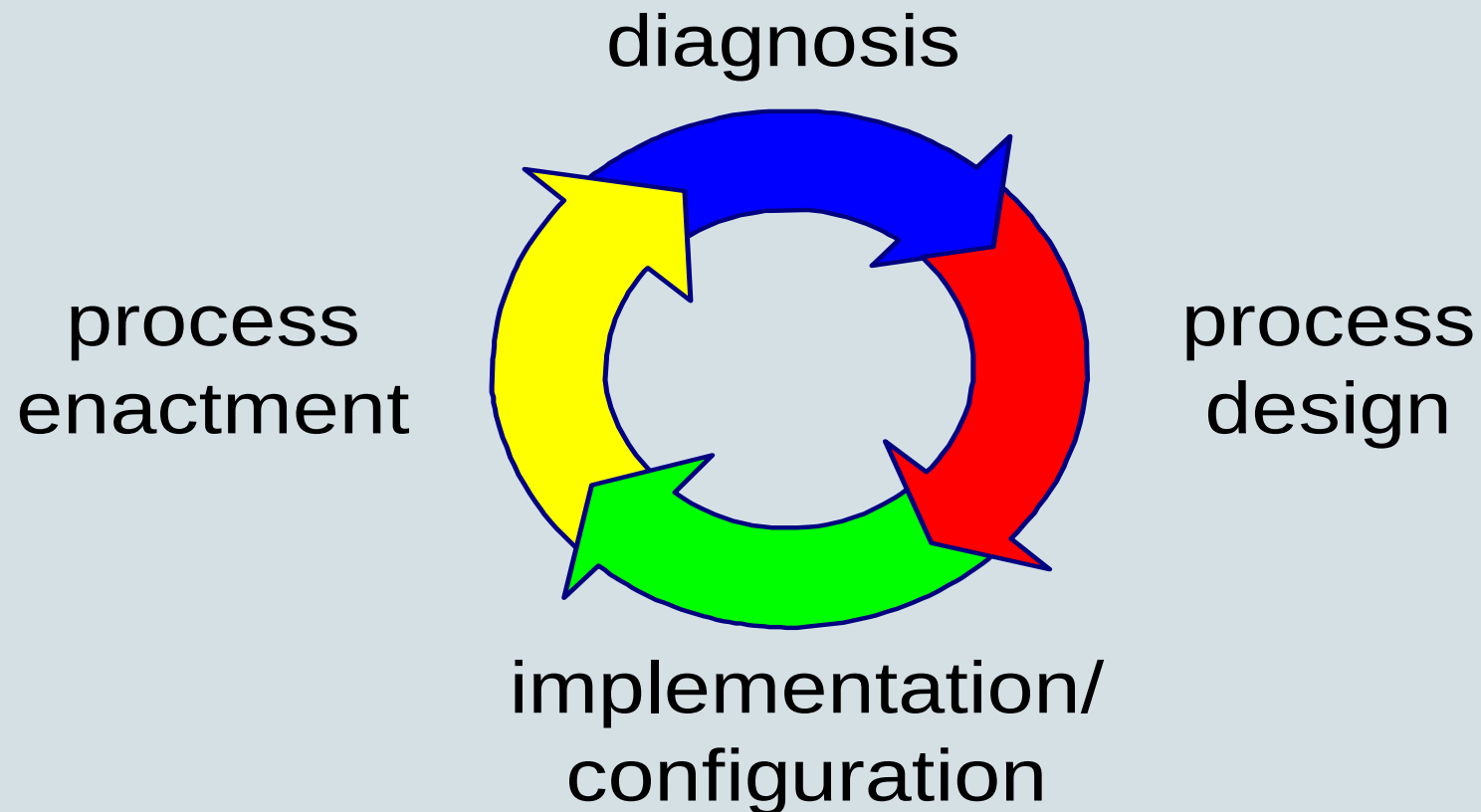
- BPM research in Eindhoven
- Process Mining
- ProM
 - Architecture
 - Process mining plug-ins
 - Alpha-algorithm
 - Multi-phase mining
 - Genetic mining
 - Analysis plug-ins
 - Conformance testing plug-in
 - LTL checker plug-in
 - Social network plug-in
 - Convertors (e-mail, Staffware, InConcert, SAP, etc.)
- Conclusion

Research within the BPM group

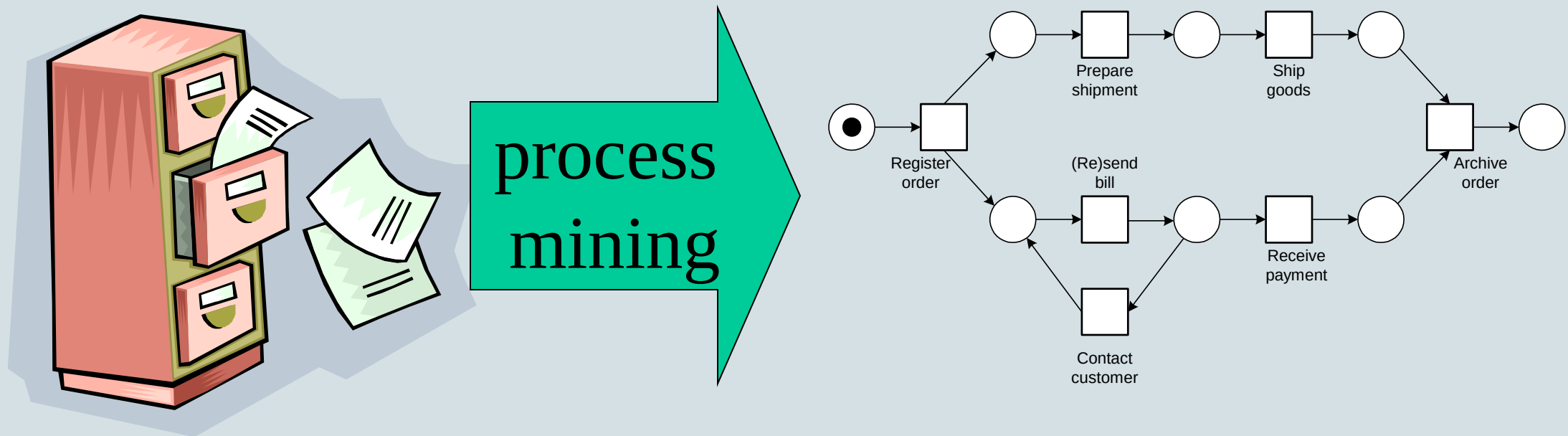
- Workflow management systems
- Process modeling (design and redesign)
- Process analysis (validation, verification, and performance analysis)
- Process mining

Often based on Petri nets

Process Mining



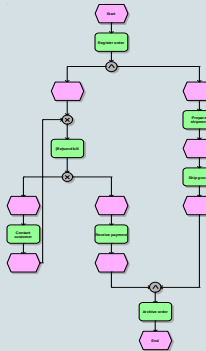
Motivation: Reversing the process



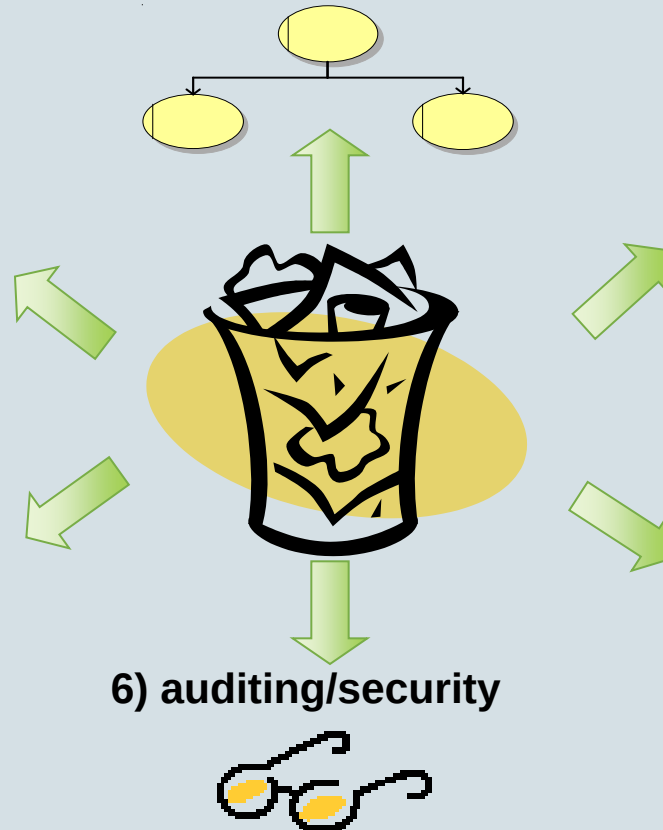
- Process mining can be used for:
 - Process discovery (What is the process?)
 - Delta analysis (Are we doing what was specified?)
 - Performance analysis (How can we improve?)

Overview

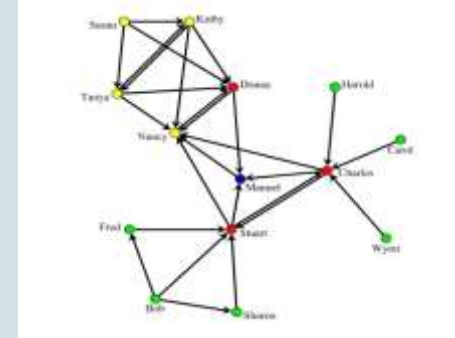
2) process model



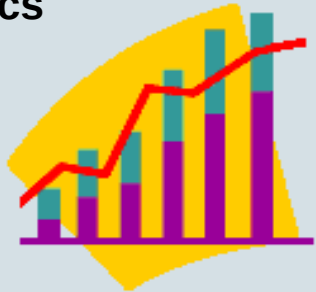
3) organizational model



4) social network



1) basic performance metrics

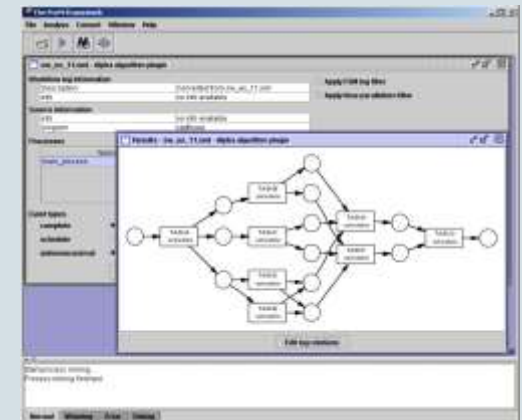


5) performance characteristics

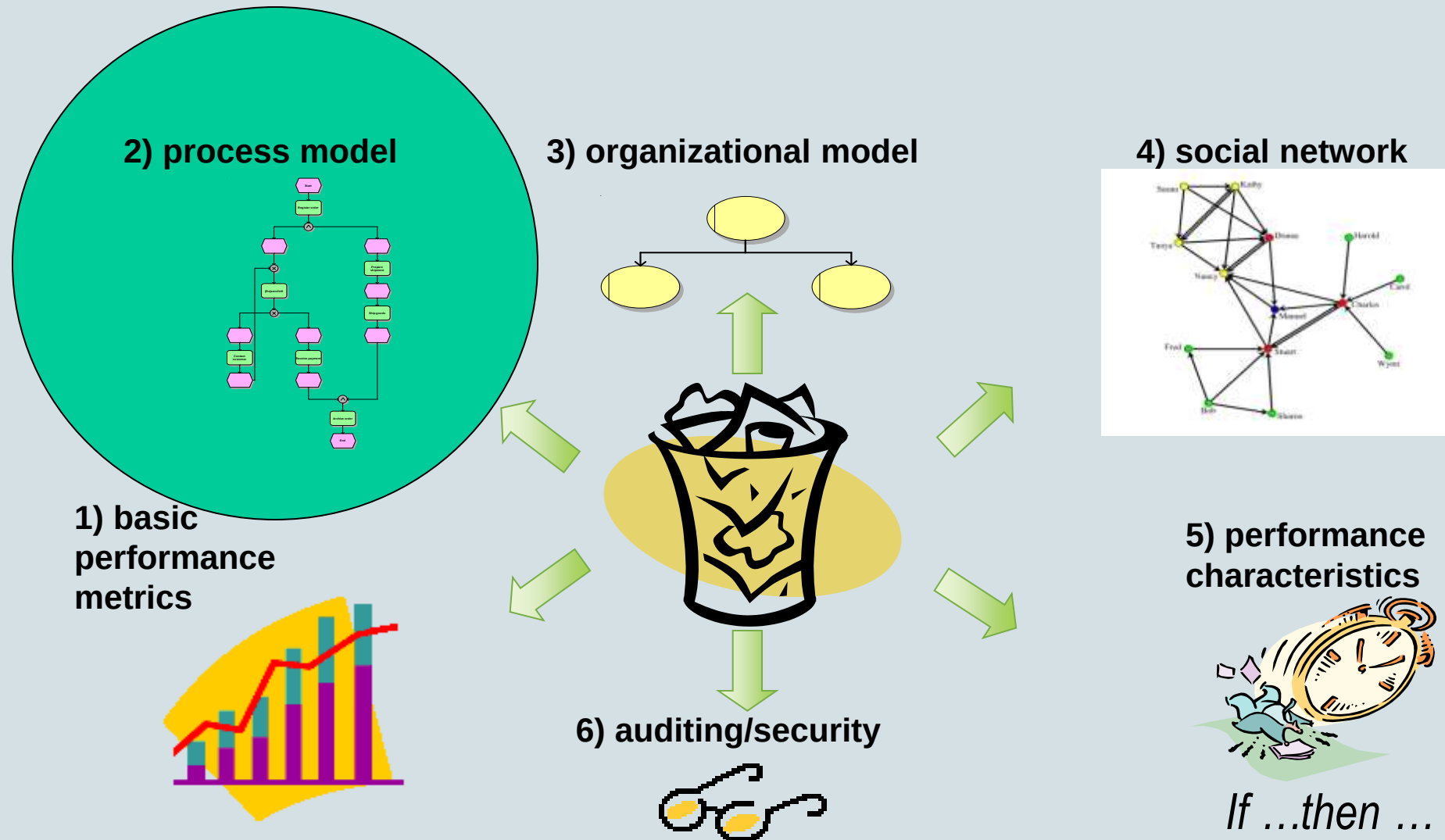


If ...then ...

www.processmining.org



Let us focus on mining process models ...

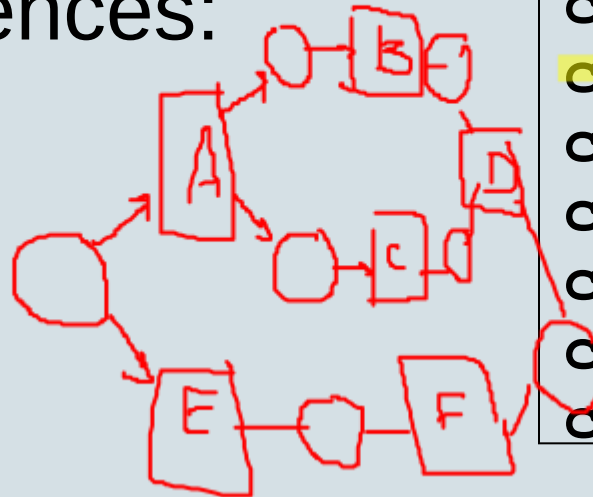


... and a very simple approach: The alpha algorithm

Process log

- Minimal information in log: case id's and task id's.
- Additional information: event type, time, resources, and data.
- In this log there are three possible sequences:

– ABCD
– ACBD
– EF



case 1	: task A
case 2	: task A
case 3	: task A
case 3	: task B
case 1	: task B
case 1	: task C
case 2	: task C
case 4	: task A
case 2	: task B
case 2	: task D
case 5	: task E
case 4	: task C
case 1	: task D
case 3	: task C
case 3	: task D
case 4	: task B
case 5	: task F
case 4	: task D

$>, \rightarrow, ||, \#$ relations

- **Direct succession:** $x > y$ iff for some case x is directly followed by y .
- **Causality:** $x \rightarrow y$ iff $x > y$ and not $y > x$.
- **Parallel:** $x || y$ iff $x > y$ and $y > x$
- **Choice:** $x \# y$ iff not $x > y$ and not $y > x$.

$A \# E$

$D \# F$

case 1	:	task A
case 2	:	task A
case 3	:	task A
case 3	:	task B
case 1	:	task B
case 1	:	task C
case 2	:	task C
case 4	:	task A
case 2	:	task B
case 2	:	task D
case 5	:	task E
case 4	:	task C
case 1	:	task D
case 3	:	task C
case 3	:	task D
case 4	:	task B
case 5	:	task F
case 4	:	task D

$A > B$

$A > C$

$B > C$

$B > D$

$C > B$

$C > D$

$E > F$

$B || C$

$C || B$

$A \rightarrow B$

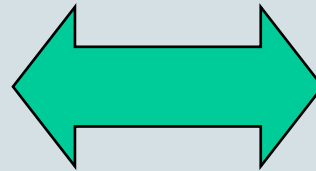
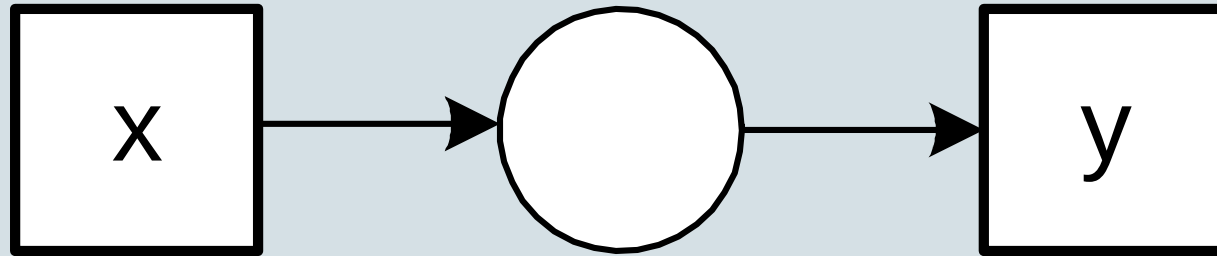
$A \rightarrow C$

$B \rightarrow D$

$C \rightarrow D$

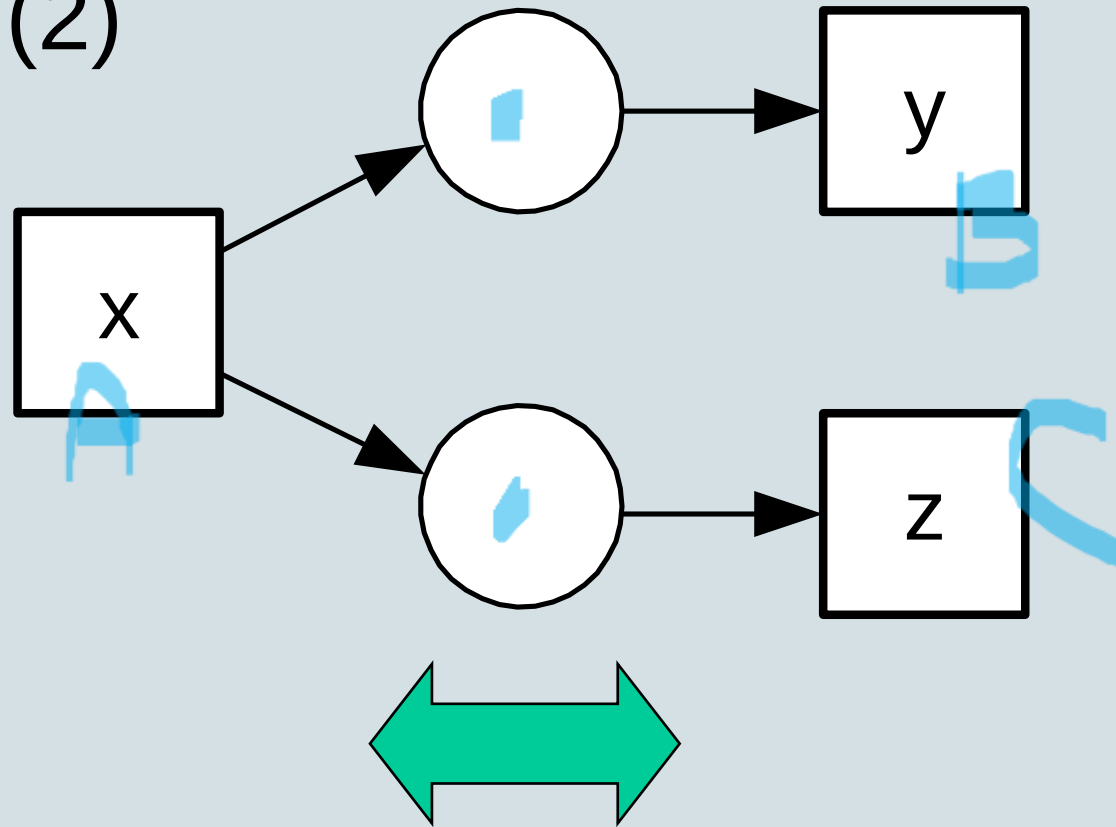
$E \rightarrow F$

Basic idea (1)



$$x \rightarrow y$$

Basic idea (2)



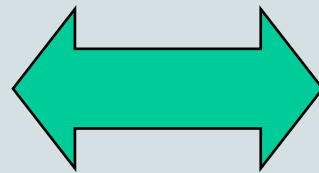
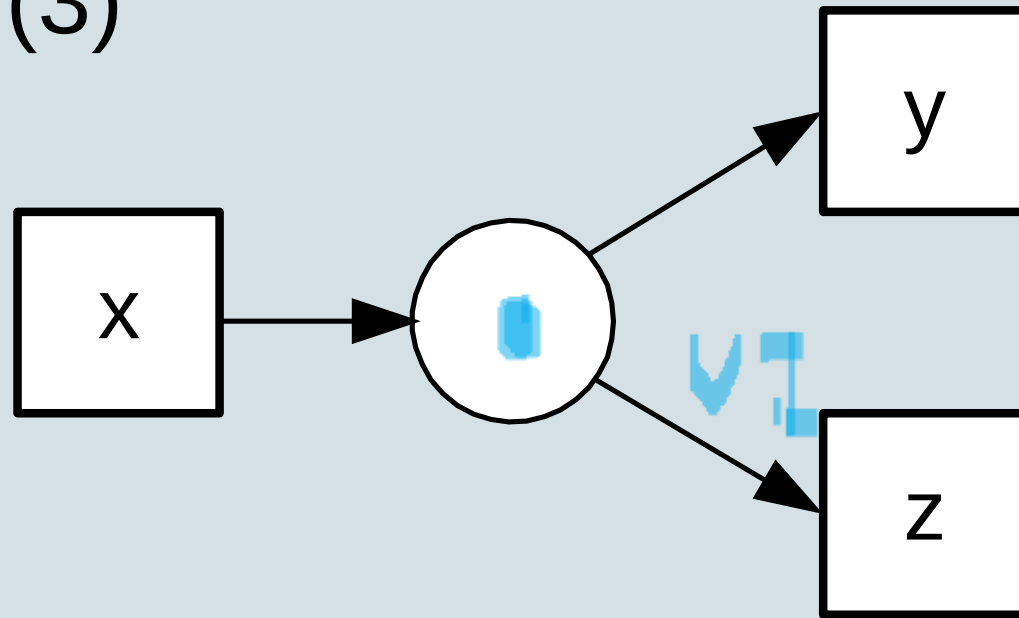
$x \rightarrow y$, $x \rightarrow z$, and $y \parallel z$

$A \rightarrow B$

$A \rightarrow C$

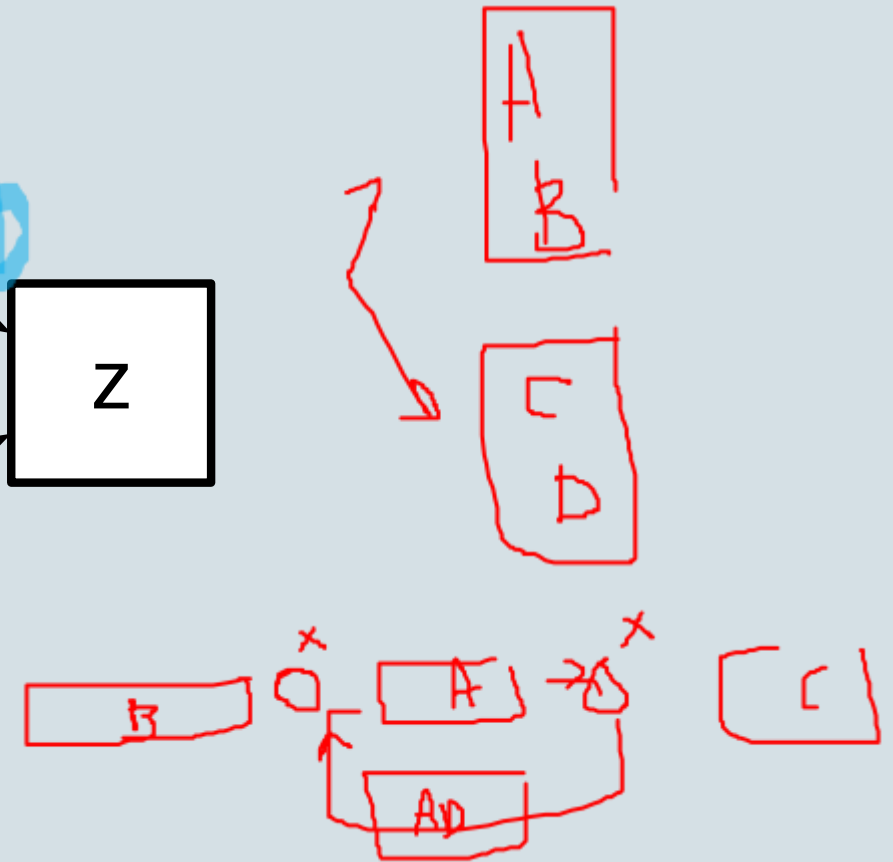
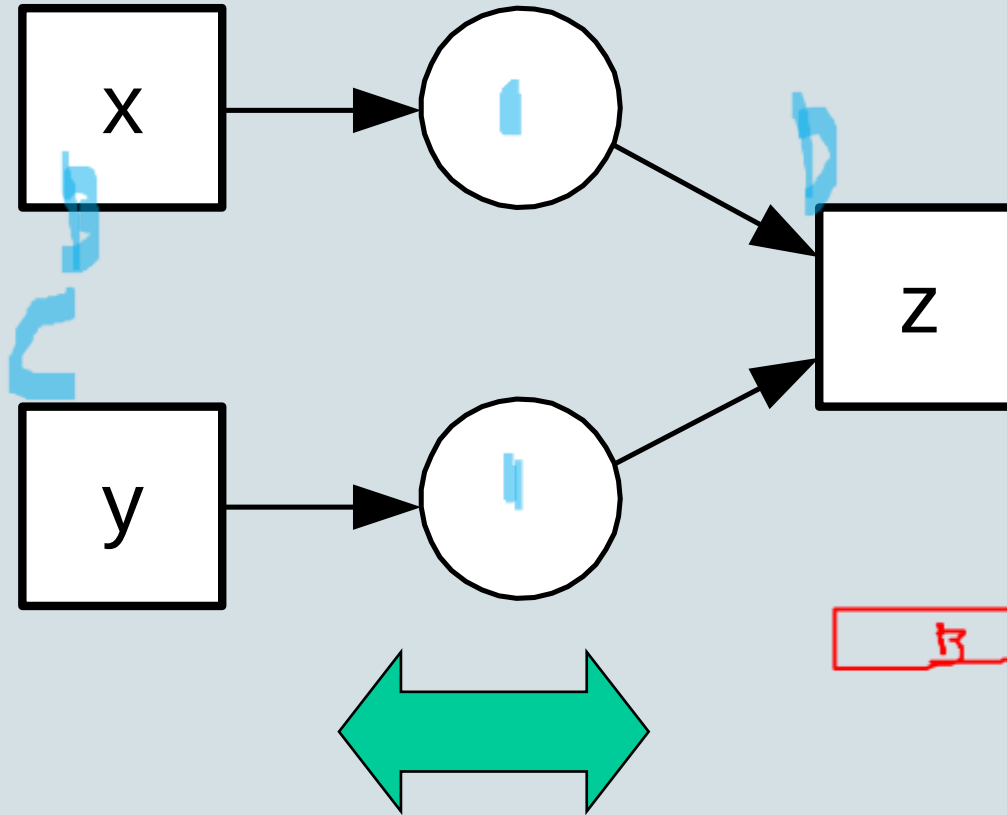
$B \parallel C$

Basic idea (3)



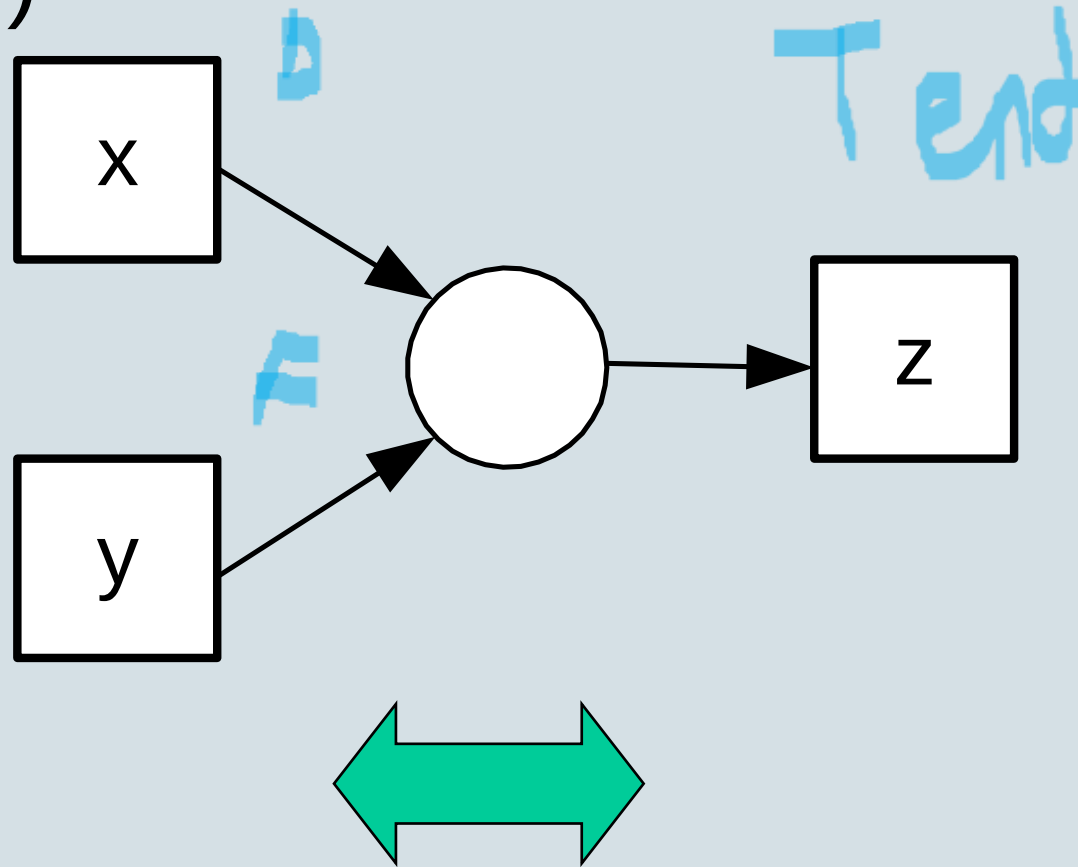
$x \rightarrow y$, $x \rightarrow z$, and $y \# z$

Basic idea (4)



$x \rightarrow z$, $y \rightarrow z$, and $x \parallel y$

Basic idea (5)



$x \rightarrow z$, $y \rightarrow z$, and $x \# y$

It is not that simple: Basic alpha algorithm

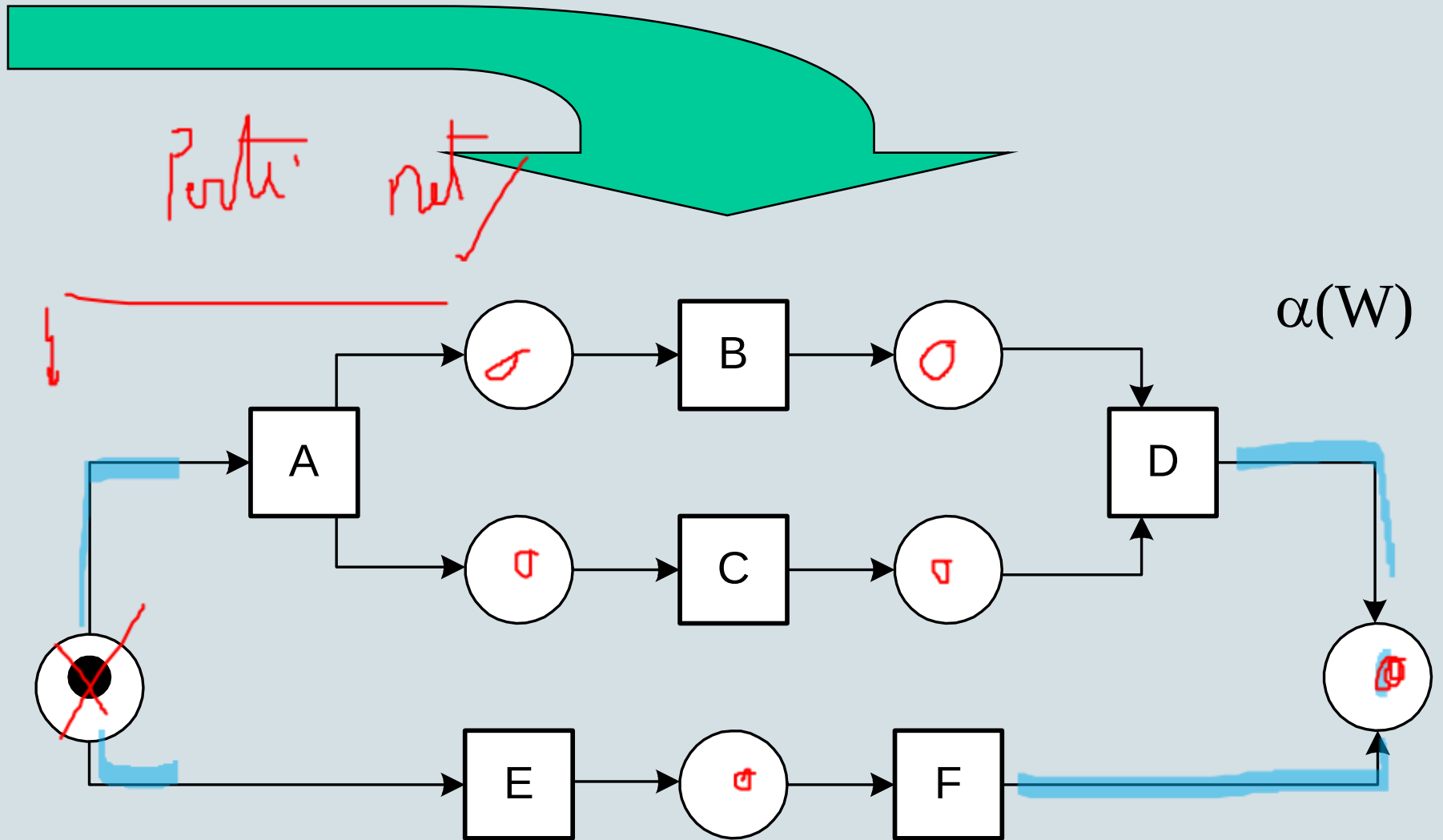
Let W be a workflow log over T . $\alpha(W)$ is defined as follows.

1. $T_W = \{ t \in T \mid \exists_{\sigma \in W} t \in \sigma \},$
2. $T_I = \{ t \in T \mid \exists_{\sigma \in W} t = \text{first}(\sigma) \},$
3. $T_O = \{ t \in T \mid \exists_{\sigma \in W} t = \text{last}(\sigma) \},$
4. $X_W = \{ (A,B) \mid A \subseteq T_W \wedge B \subseteq T_W \wedge \forall_{a \in A} \forall_{b \in B} a \rightarrow_W b \wedge \forall_{a_1, a_2 \in A} a_1 \#_W a_2 \wedge \forall_{b_1, b_2 \in B} b_1 \#_W b_2 \},$
5. $Y_W = \{ (A,B) \in X \mid \forall_{(A',B') \in X} A \subseteq A' \wedge B \subseteq B' \Rightarrow (A,B) = (A',B') \},$
6. $P_W = \{ p_{(A,B)} \mid (A,B) \in Y_W \} \cup \{ i_W, o_W \},$
7. $F_W = \{ (a, p_{(A,B)}) \mid (A,B) \in Y_W \wedge a \in A \} \cup \{ (p_{(A,B)}, b) \mid (A,B) \in Y_W \wedge b \in B \} \cup \{ (i_W, t) \mid t \in T_I \} \cup \{ (t, o_W) \mid t \in T_O \},$ and
8. $\alpha(W) = (P_W, T_W, F_W).$

The alpha algorithm has been proven to be correct for a large class of free-choice nets.

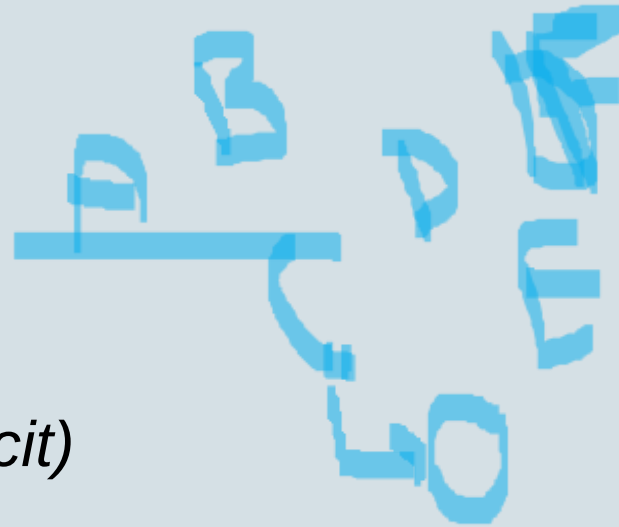
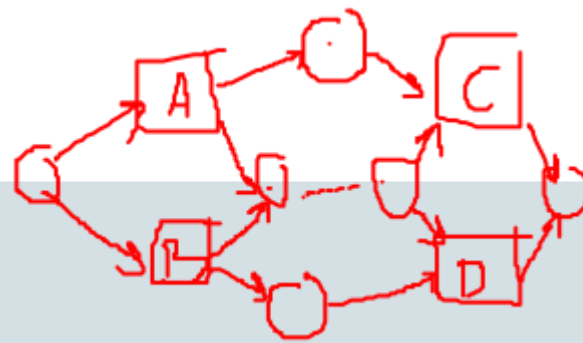
w Example

case 1 : task A
case 2 : task A
case 3 : task A
case 3 : task B
case 1 : task B
case 1 : task C
case 2 : task C
case 4 : task A
case 2 : task B
case 2 : task D
case 5 : task E
case 4 : task C
case 1 : task D
case 3 : task C
case 3 : task D
case 4 : task B
case 5 : task F
case 4 : task D



Challenges

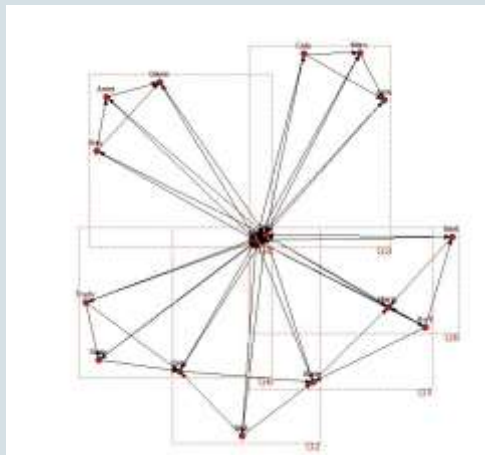
- Refining existing algorithm for (control-flow/process perspective)
 - *Hidden tasks*
 - *Duplicate tasks*
 - *Non-free-choice constructs*
 - *Loops*
 - *Detecting concurrency (implicit or explicit)*
 - *Mining and exploiting time*
 - *Dealing with noise*
 - *Dealing with incompleteness*
- Mining other perspectives (data, resources, roles, ...)
- Gathering data from heterogeneous sources
- Visualization of results
- Delta analysis



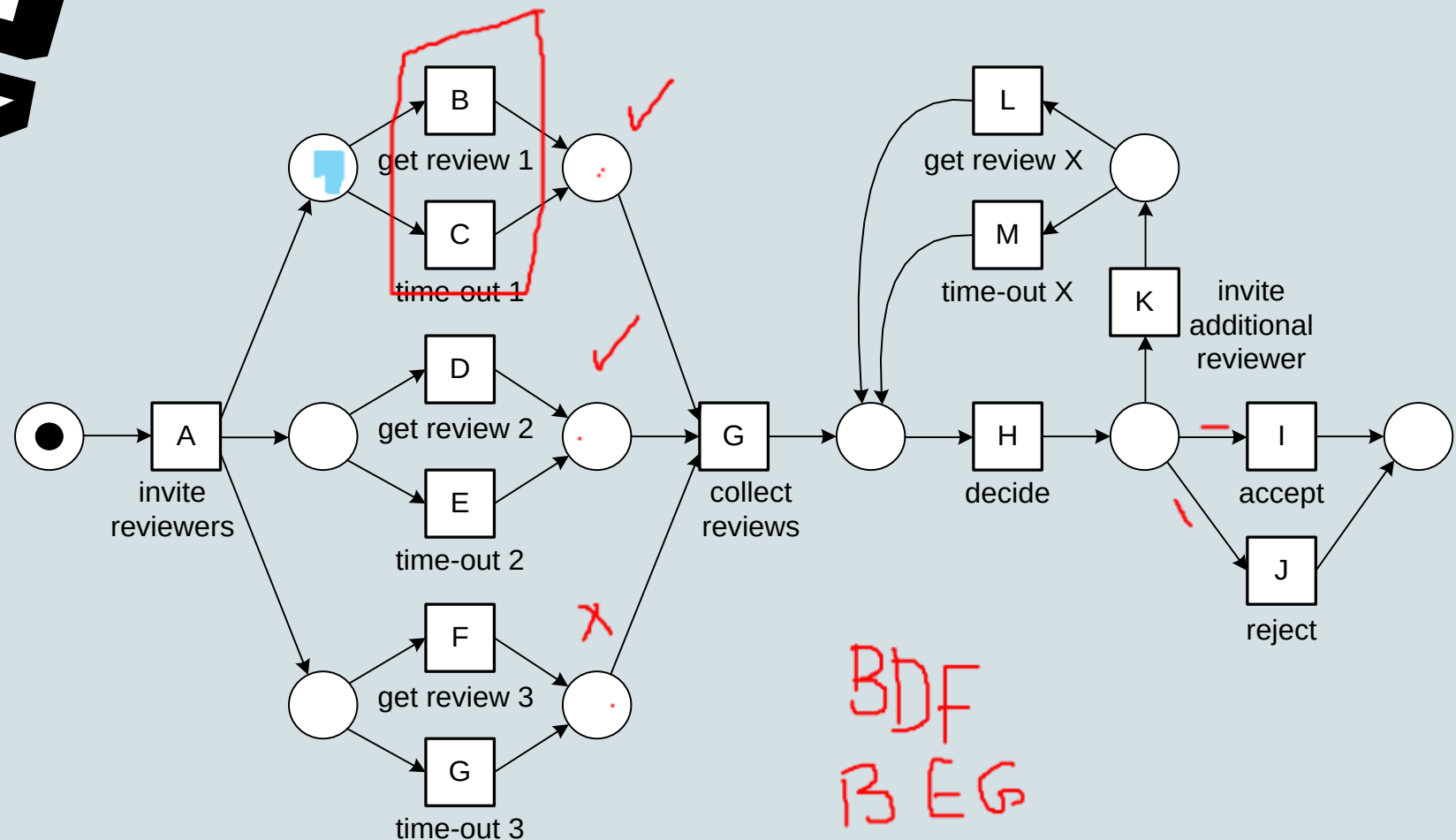


DEMO

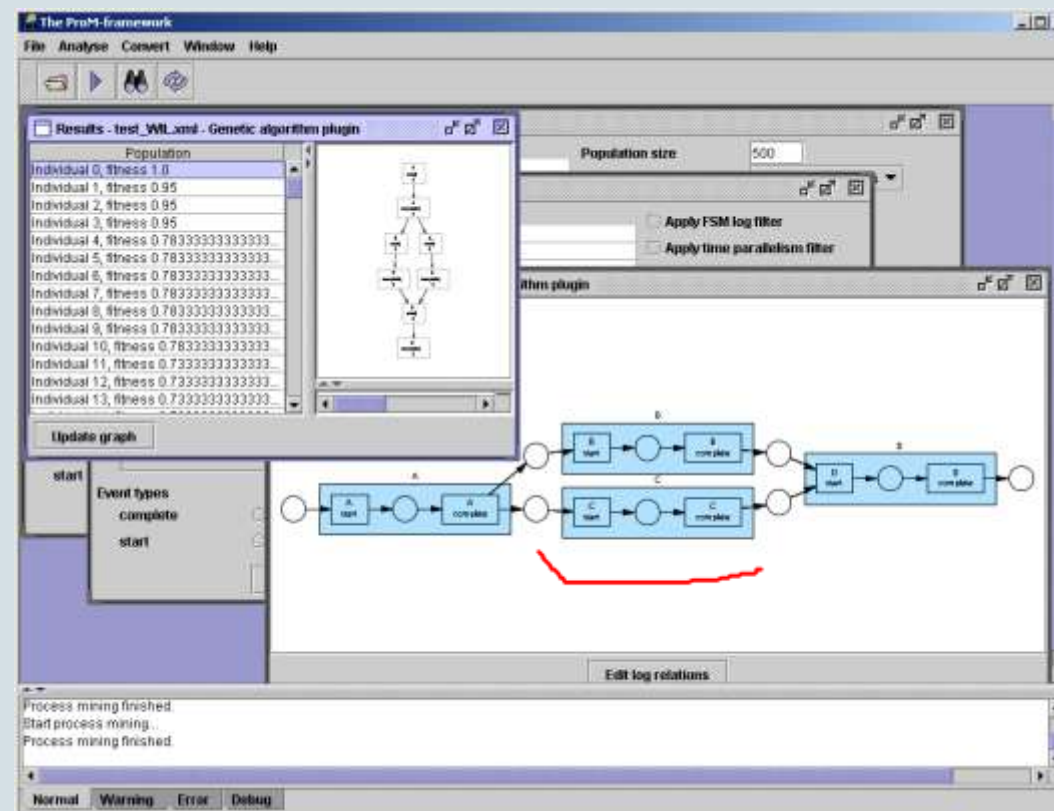
Alpha algorithm



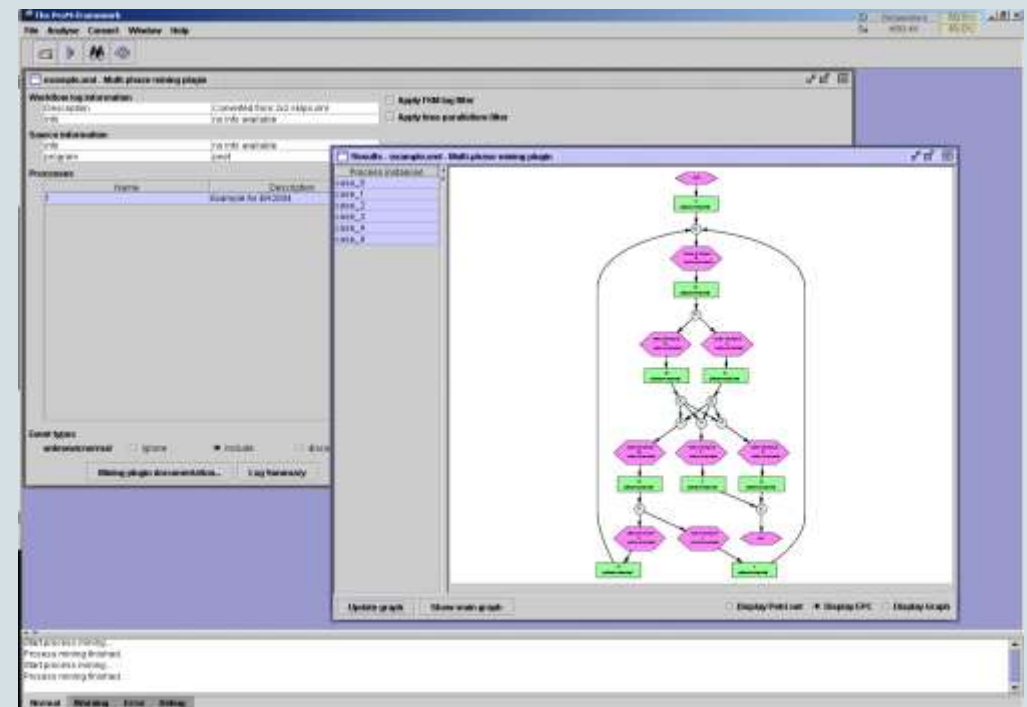
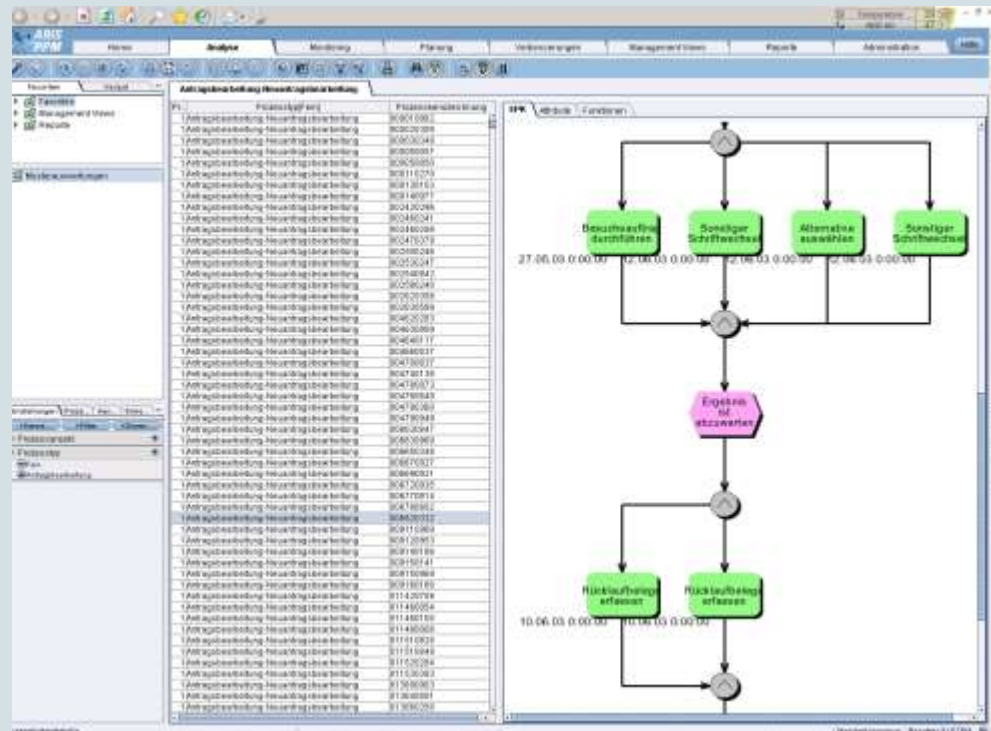
48 cases
16 performers



ProM framework

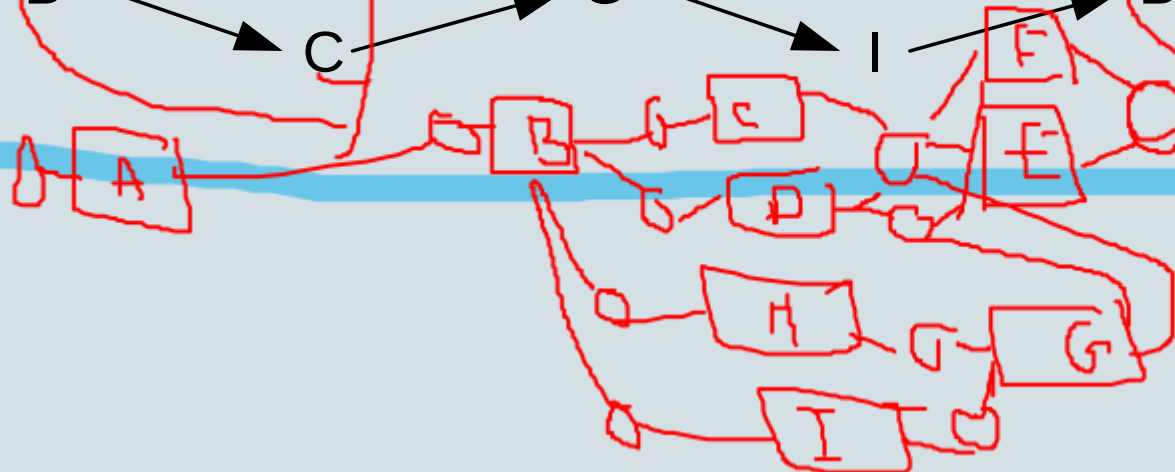
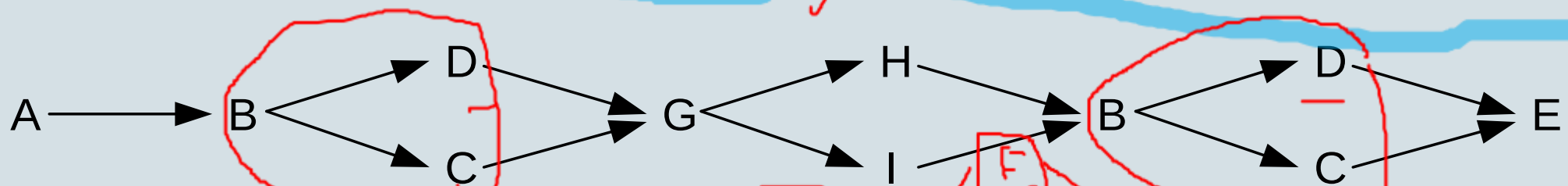
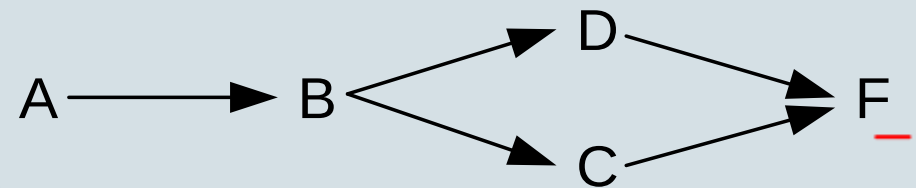
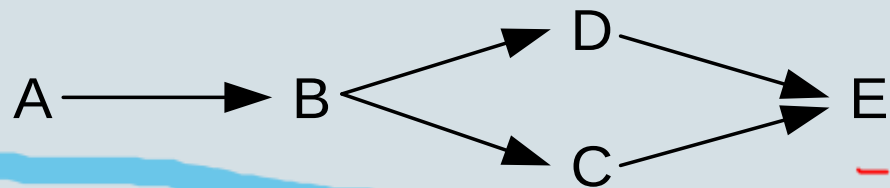


Mining plug-in: Multi-phase mining

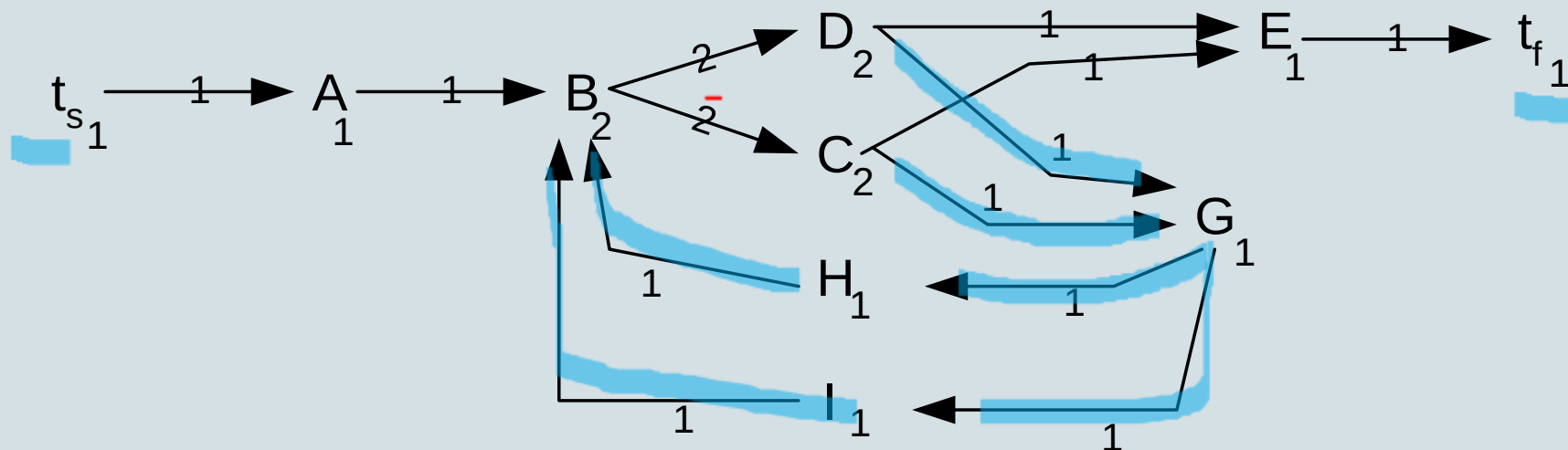
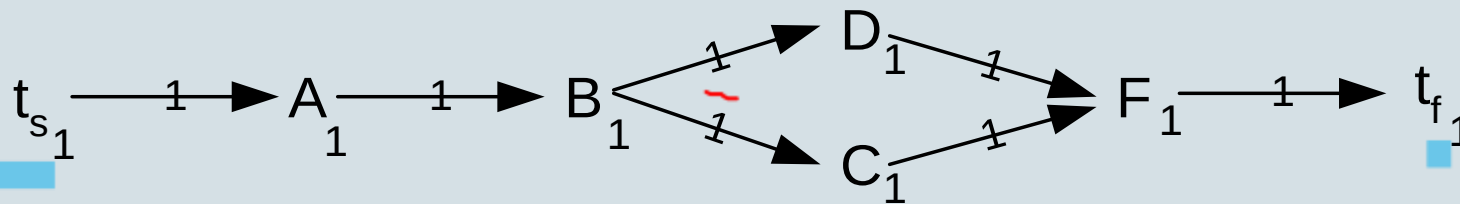
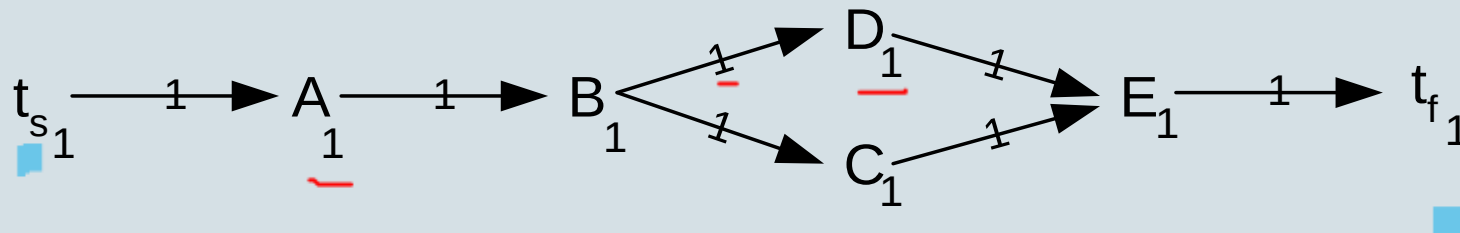


Step 1: Get instances

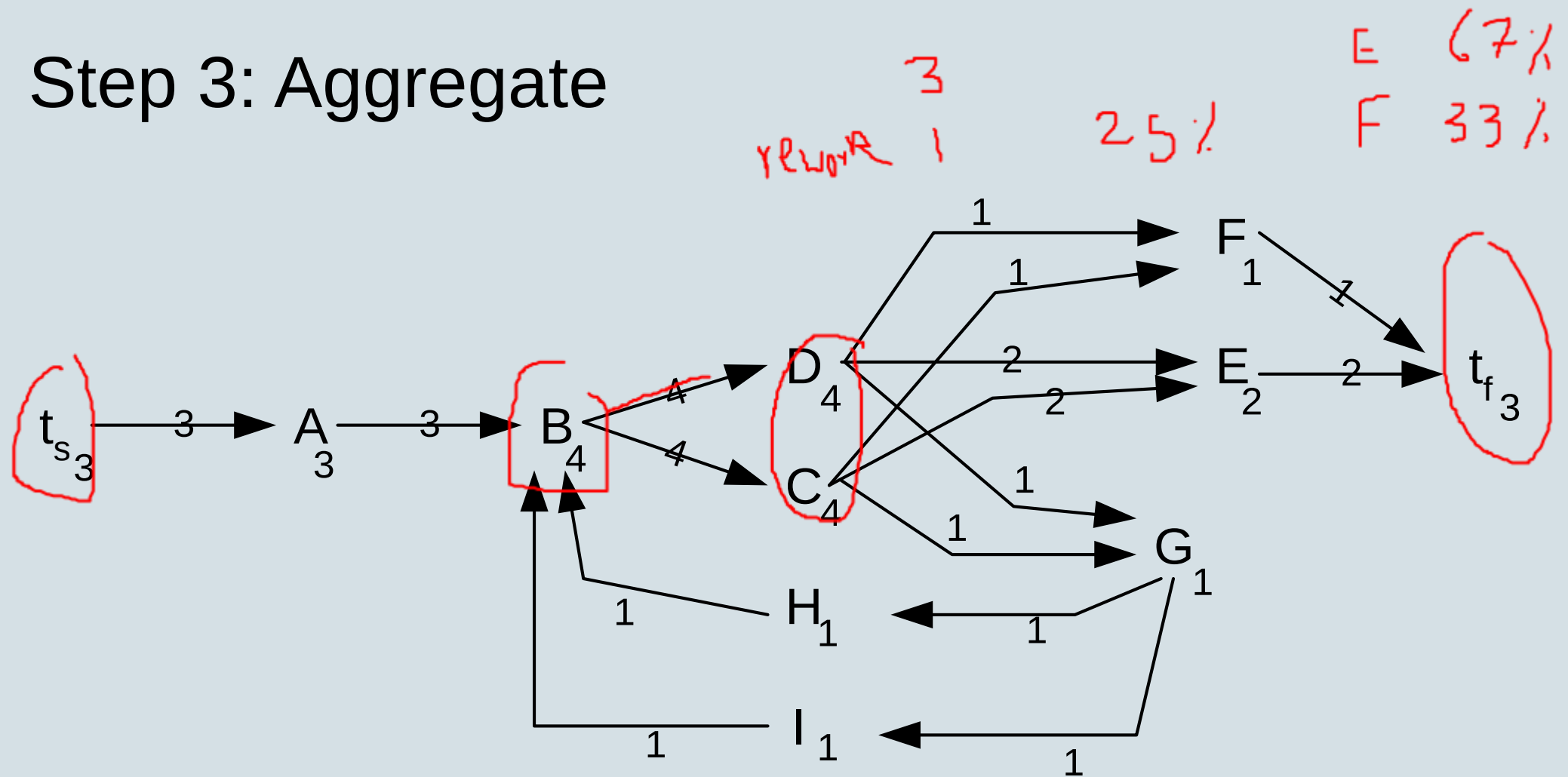
No Loop
rework



Step 2: Project

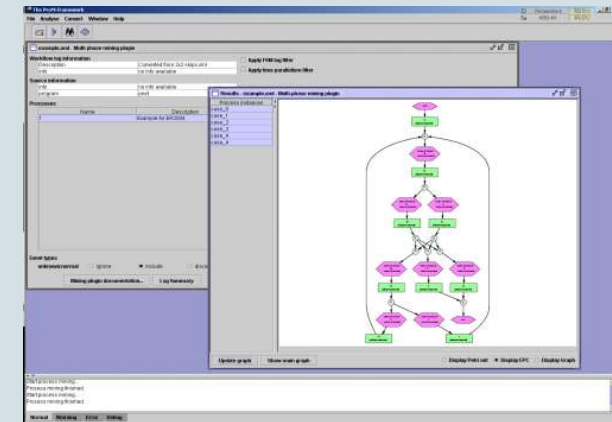
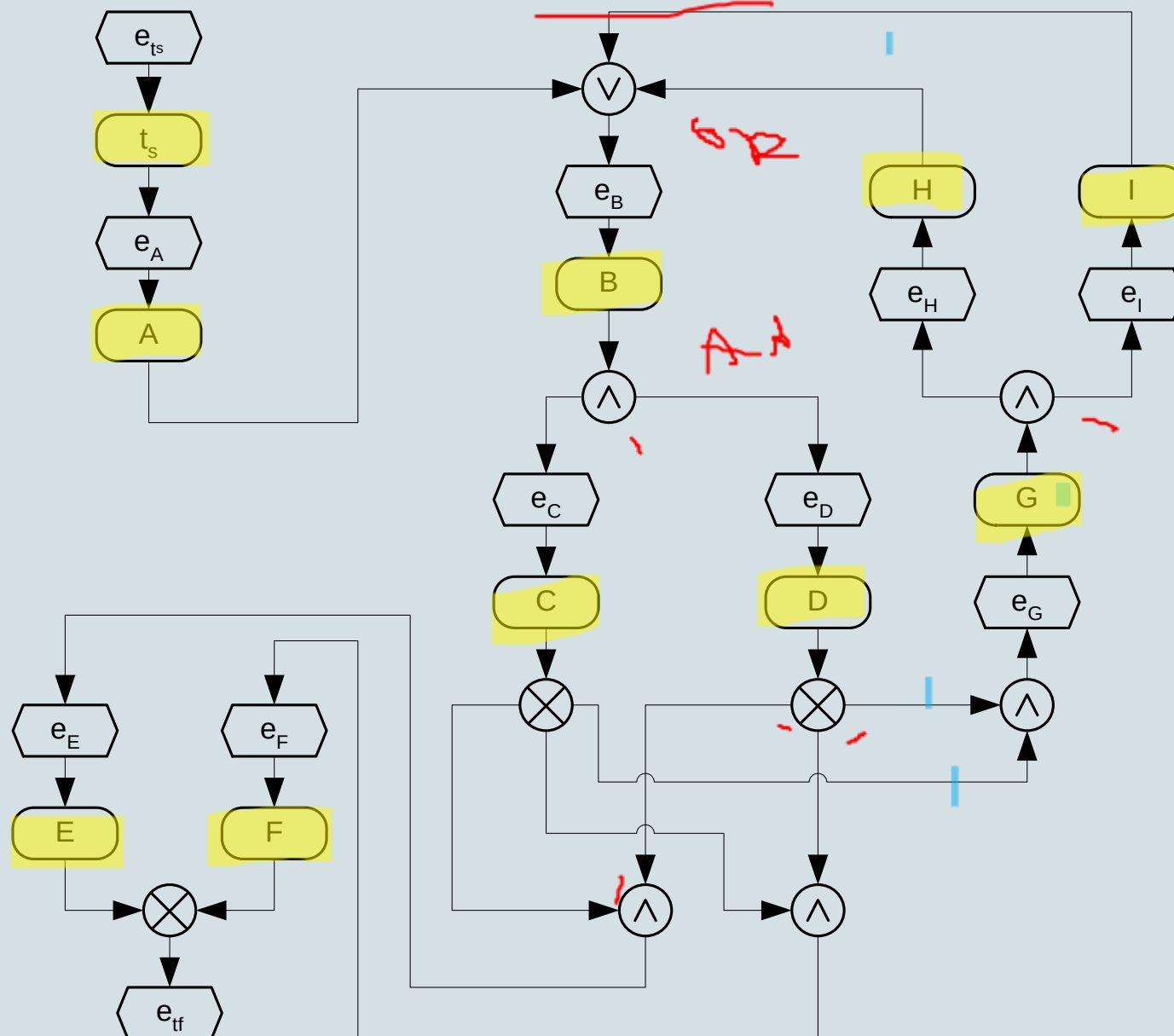


Step 3: Aggregate

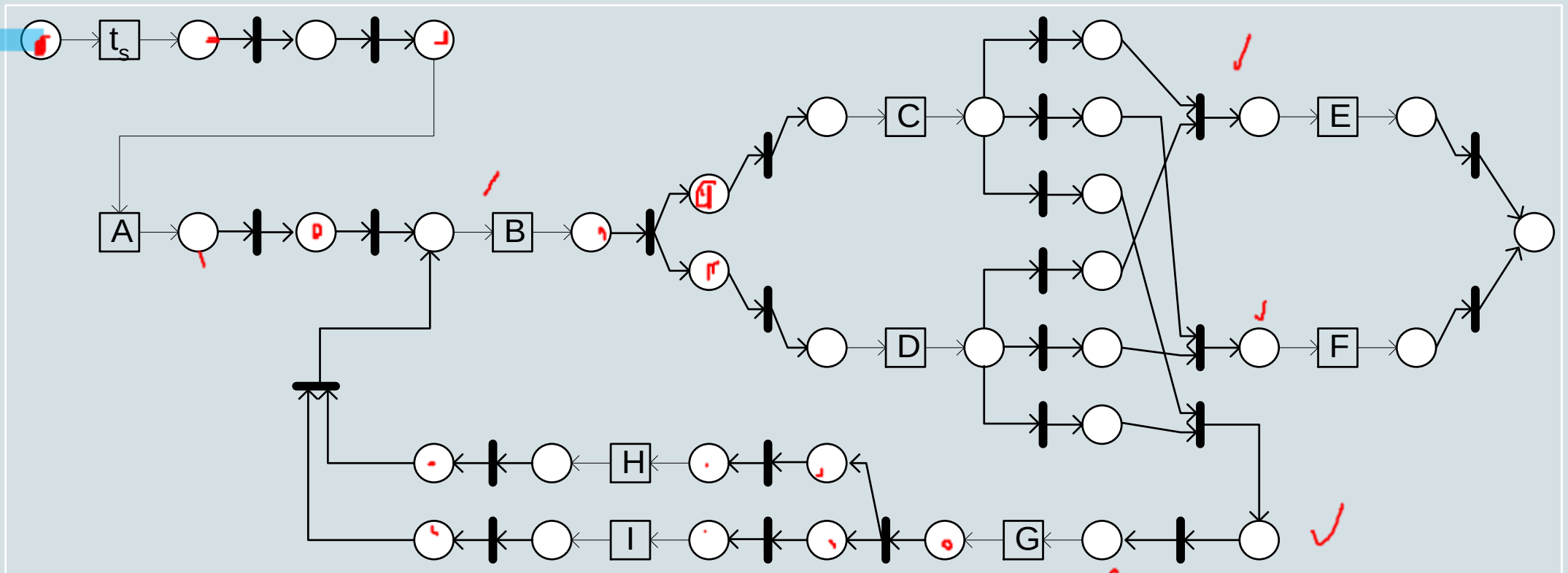


EPC: event-driven process chain diagram

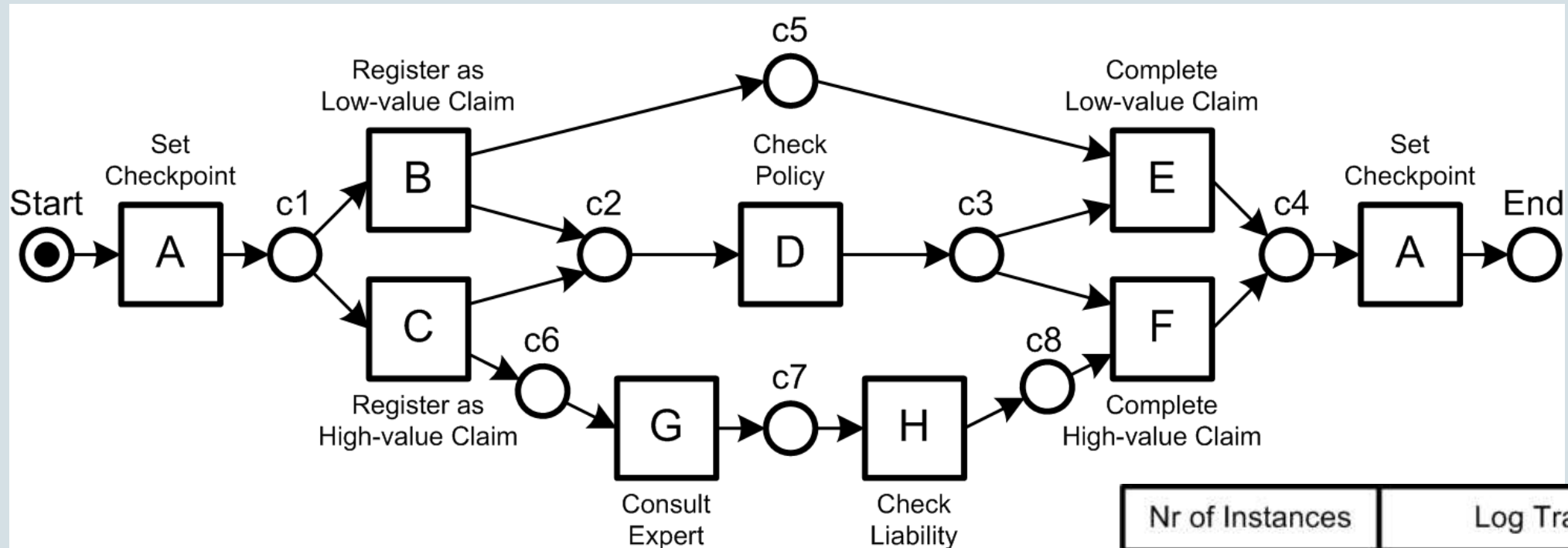
Step 4: Map onto EPC



Step 5: Map onto Petri net (or other language)

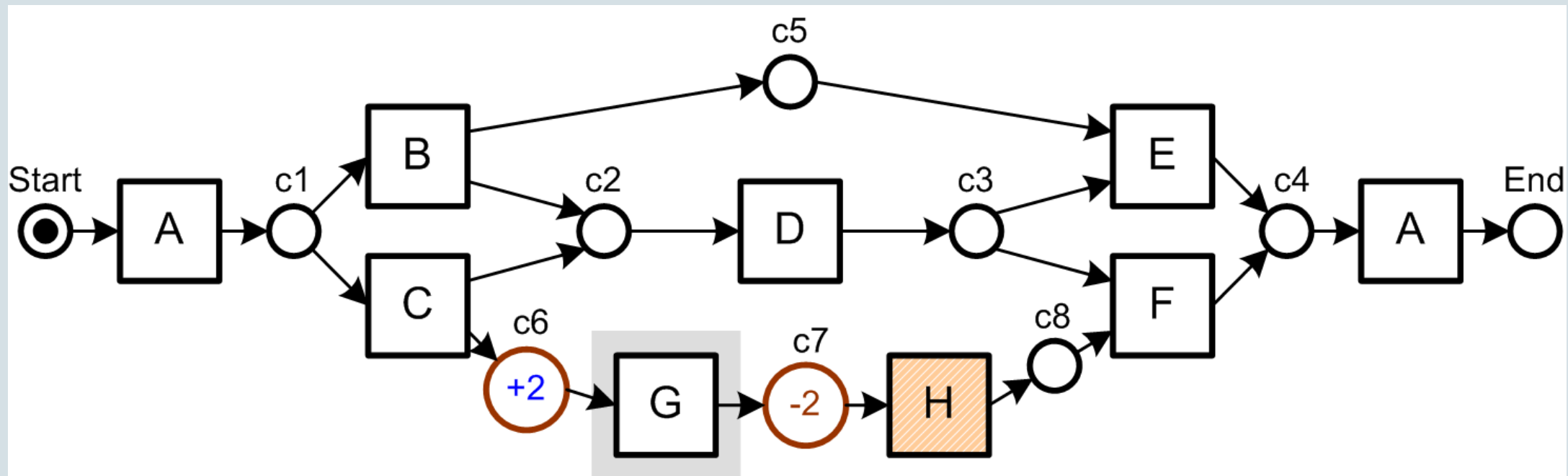


Analysis plug-in: Conformance checker



Do they agree?

Nr of Instances	Log Traces
1207	ABDEA
145	ACDGHFA
56	ACGDHFA
23	ACHDFA
28	ACDHFA



Nr of Instances	Log Traces
1207	ABDEA
145	ACDGHFA
56	ACGDHFA
23	ACHDFA
28	ACDHFA

Screenshot

The ProM-framework

File Mining Analysis Conversion Exports Window Help

Imported - performance.tpn

Analysis - Conformance Checker

#	Log Traces
1207	0
145	1
56	2
23	3
28	4

Model Related Measures

Fitness: 1.0

Appropriateness: 0.97057164

Analysis - Conformance Checker

#	Log Traces
1207	0
145	1
56	2
23	3
28	4

Model Related Measures

Fitness: 0.99446076

Appropriateness: 0.970471

Update Results

Diagnostic Perspective

Model

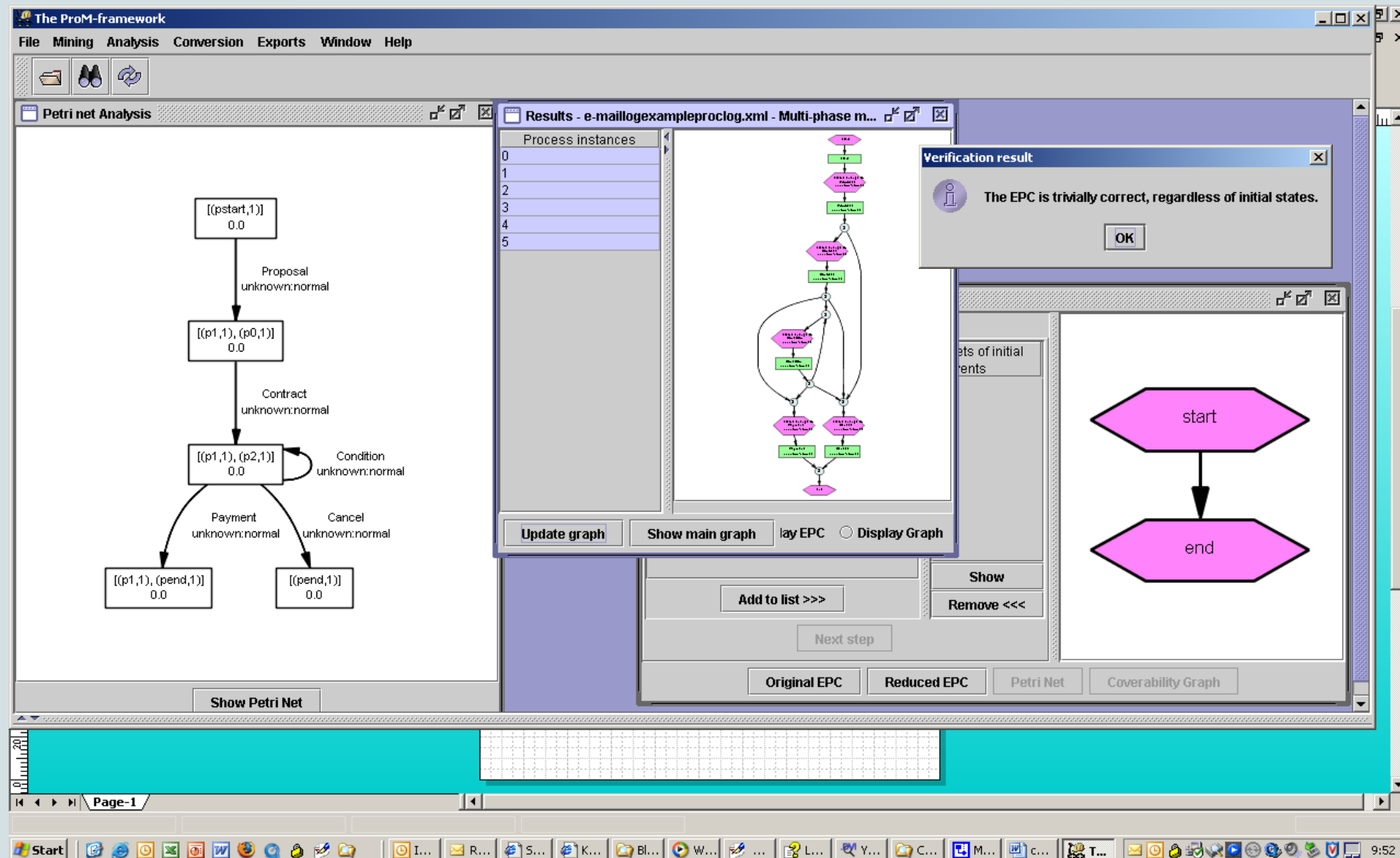
☒ Token counter ☒ Failed Tasks ☒ Remaining Tasks ☐ Path coverage

finished initializing.
Please wait while initializing the Log....
finished initializing.
/Users/anne/Desktop/demo/example files/nonFitting.tpn imported.

Normal Warning Error Debug

(Also runs on Mac.)

Other analysis plug-ins



Conclusion

- Process mining provides many interesting challenges for scientists, customers, users, managers, consultants, and tool developers.
- Involves multiple perspectives (process, data, resources, etc.)
- **Get ProM-ed!**
- You can contribute by applying ProM and developing plug-ins

More information

<http://www.processmining.org>